

Bhuvanesh Bhatt

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[LinkedIn](#) · [GitHub](#)

Software engineer with 25 years of experience in scientific computing, machine learning systems, and large-scale symbolic and numeric computation, spanning both implementation and testing. Extensive experience with ML pipelines and LLM APIs. Developed mathematical functionality including stochastic-process models and integration methods. Built validation infrastructure that expanded automated coverage by tens of thousands of tests, discovered thousands of defects, reduced unexplained failures substantially, and improved the reliability of complex computational systems across data science, symbolic and numeric math, code compilation and data frameworks. Seeking to apply expertise in data science and computational mathematics to mission-driven or research-oriented teams.

Core Machine Learning & Data Science Expertise

- Supervised & unsupervised learning:
Classification, regression, ensemble methods (gradient boosting, random forests), clustering, anomaly detection, missing value imputation, dimensionality reduction, regularization, neural networks
- ML data pipeline validation:
Feature engineering, preprocessing, hyperparameter tuning, model evaluation including cross-validation
- Model interpretability & explainability:
SHAP value validation via axiomatic verification (efficiency, symmetry, dummy, linearity), perturbation faithfulness, approximation stability, attribution consistency, and synthetic ground-truth benchmarks
- LLM & agentic AI systems integration:
Prompt templating, tool-calling, multi-step agent pipelines, multimodal generation, response validation
- ML reliability & regression testing:
Deterministic behavior, numerical stability, performance benchmarking, cross-platform reproducibility
- Exploratory analysis & reporting:
Prototyping and building models to answer questions, visualizing data and communicating results

Core Scientific Computing & Numerical Methods Expertise

- Computational mathematics:
Linear algebra, statistics, optimization, graph theory, geometry, differential geometry, tensor algebra, calculus, commutative and noncommutative algebra, equation and inequality solving, special functions
- Statistical inference:
Parametric and nonparametric hypothesis testing, confidence intervals, resampling methods, correlation analysis, and distributional comparison; Bayesian inference; uncertainty analysis & sensitivity analysis
- Numerical modeling & simulation:
Regression, generalized linear models, hierarchical/mixed-effects models, time-series modeling and forecasting, Monte Carlo methods including MCMC
- Scientific software quality and reproducibility:
Reproducible workflows, structured data validation, cross-platform consistency testing, benchmark-driven performance analysis, and large-scale automated property-based test design

Programming and Frameworks

- Python (machine learning, scientific computing, geospatial libraries), Mathematica (Wolfram Language), C/C++, Java, Kotlin, R
- MySQL, PostgreSQL, MongoDB, Neo4j
- PyTorch, TensorFlow
- Apache Kafka, Spark, Airflow, dbt
- Javascript, Django, React, Node

Tools and Platforms

Git, CMake, dotenv, uv, Ruff, pyright/pylance, pytest, Docker, Kubernetes, TeamCity, AWS, QGIS, Tableau

Professional Experience

Senior Software Quality Assurance Engineer | August 2014 – February 2026

[Wolfram Research](#)

- Tested classical machine learning functionality, automatic preprocessing pipelines (type inference, categorical encoding, missing value handling), interpretability, and end-to-end training pipelines
- Verified large language model (LLM) API and agentic AI integrations, including prompt templating, chat sessions, tool-calling workflows, multi-step agents, and multimodal generation

- Conducted white- and black-box testing, regression and performance testing of symbolic computation (~140k tests), special functions (~16k tests), machine learning (~20k tests)
- Developed tooling to infer valid Mathematica function argument structures and generate mathematically meaningful test inputs at scale, fed those to sanity-checking and property-based testing utilities, replacing largely ad hoc testing practices with domain-aware input generation and exposing branch-cut issues, precision problems, and incorrect evaluations/simplifications
- Expanded automated regression coverage by a few thousand tests through tooling that converted non-graphics documentation examples into executable tests aligned with user-facing behavior
- Improved performance benchmarking reliability by replacing fixed, fragile timing thresholds with machine- and load-aware analysis, making true slowdowns easier to distinguish from noise
- Reduced unexplained failure rates 8% to under 2% via systematic cleanup and root-cause analysis
- Identified, isolated, and verified resolution of thousands of defects in computational algorithms
- Validated the LLVM-based Wolfram Compiler infrastructure, verifying correctness, performance improvements, and numerical consistency between interpreted and compiled execution paths
- Validated curated datasets including geospatial, financial, chemical, astronomical & weather data
- Also validated computational geometry and graph theory functionality
- Was frequently pulled into reviews of mathematical changes outside my formal areas, serving as a trusted reviewer for computational functionality
- Gave a Wolfram Tech Conference talk on quality assurance processes for Mathematica

Software Developer | October 2011 – July 2014

[Wolfram Research](#)

- Designed and implemented support for Tsallis q-Gaussian and q-exponential distributions in Mathematica's probability and statistics functionality
- Developed functionality for discrete-time, continuous-time, and hidden Markov models, including profiling, benchmarking and optimizing for efficiency
- Implemented the layer-cake method for multiple integrals, extending Mathematica's symbolic integration capabilities for structured integral problems
- Implemented harmonic polylogarithms, multiple polylogarithms, and the multiple zeta function
- Researched practical applications and created documentation examples that helped connect advanced statistical functionality to real user workflows, with visualizations
- Collaborated with cross-functional teams including quality assurance and tech documentation
- Gave Tech Conference talks on Mathematica with an emphasis on statistics and parallel computing

SQA Engineer | February 2001 – September 2011

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- Tested the symbolics and special functions components
- Validated cross-language APIs connecting Mathematica with C/C++, Java and .NET environments
- Used gcov coverage reports to identify untested code paths and drive systematic test design
- Designed large-scale correctness, regression, and benchmark validation infrastructure

Education and Research

- **M.S. Computer Science** (incomplete) — GPA 3.83 · Towson University
 - Thesis: Survey of middleware solutions
 - Courses included data structures & algorithms (using C++/Java), computer architecture, operating systems, middleware, networking, database systems, software engineering, simulation & modeling
- **B.S. Physics, minor in Mathematics** — GPA 3.74 · Towson University
 - Undergraduate research in numerical simulation (radiative transfer modeling in Fortran) and scientific instrumentation (DSP interfacing for AFM, LabView)

Professional References

Available upon request.